(1) The turnout of spectators at the world cup cricket match is independent upon the weather. On a rainy day the probability of a big turnout is 0.3. If it doesn't rain, then the probability of big turnout increases by 0.6. The weather forecast gives a probability of 0.75 that it will rain on the day of the match.

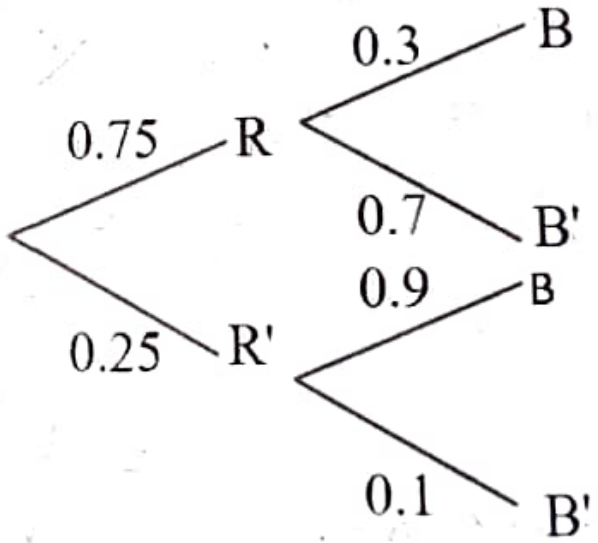
Find the probability that (i) There is a big turn out and it rains (ii) There is a big turnout.

Solution : Let R = It rains,

R' = It doesn't rain

B = Big turnout,

B = No big turnout.



$P(R) = 0.75, \quad P(R') = 0.25$

$P(B/R) = 0.3$

$P(B/R') = 0.9$

(i) $P(B \cap R) = P(B/R).P(R)$

$= 0.3 \times 0.75 = 0.225$

(ii) $P(B) = P(B/R).P(R) + P(B/R').P(R')$

$= 0.3 \times 0.75 + 0.25 \times 0.9 = 0.45$

(2) A bag contains 7 red and 5 blue balls. A ball is taken at random from the bag, its color is noted and not replaced in to the bag. Now a second ball is taken from the bag and its color is noted. Find the probability that one is red and the other is blue.

Solution : Let A be the event that in the first draw we get a red ball.

$\therefore$ A' is the event that in the first draw we do not get a red ball (i.e. we get blue ball)

Let B be the event that in the second draw we get a red ball.

$\therefore$ B' is the event that in the second draw we do not get a red ball (i.e. we get blue ball)

To find $P\left[(A\cap B')\cup(A'\cap B)\right]$

$$P\left[(A\cap B')\cup(A'\cap B)\right] = P(A\cap B')+P(A'\cap B)$$

$$=P(A)P(B'/A)+P(A')P(B/A')$$

$$= \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{70}{132} = \frac{35}{66}$$

(3) If $P(A) = \frac{1}{4}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) = \frac{1}{2}$ then find :

1) $P(A \cap B)$ 2) $P(A \cap B')$ 3) $P(A' \cap B)$ 4) $P(A' \cup B')$ 5) $P(A' \cap B')$

Solution : We know that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\therefore \frac{1}{2} = \frac{1}{4} + \frac{2}{5} - P(A \cap B)$

$\therefore P(A \cap B) = \frac{1}{4} + \frac{2}{5} - \frac{1}{2} = \frac{5+8-10}{20} = \frac{13}{20} = \frac{3}{20}$

$P(A') = 1 - P(A) = \frac{3}{4}$. and $P(B') = 1 - P(B) = 1 - \frac{2}{5} = \frac{3}{5}$

1) $P(A \cap B) = \frac{3}{20}$

2) $P(A \cap B') = P(A) - P(A \cap B) = \frac{1}{4} - \frac{3}{20} = \frac{5-3}{20} = \frac{2}{20} = \frac{1}{10}$

3) $P(A' \cap B) = P(B) - P(A \cap B) = \frac{2}{5} - \frac{3}{20} = \frac{5}{20} = \frac{1}{4}$

4) $P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B) = 1 - \frac{3}{20} = \frac{17}{20}$

5) $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B) = 1 - \frac{1}{2} = 1/2$

$1 - \frac{1}{2} = 1/2$

(4) A coin is tossed until a head appears or until it has been tossed three times. Given that head does not occur on the first toss, what is the probability that coin is tossed three times ?

Solution : Since coin is tossed three times

$\therefore S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Given that head does not occurs in the first toss.

$\therefore$ New sample will be

$S = \{THH, THT, TTH, TTT\}$

In the following cases

1. HH
2. HT
3. TH
4. TT

Now out of these 4 cases we will be required to toss coin thrice if the situation fails under

① case 3 (TH) & ② case 4 (TT)

we get head at 3rd ~~class~~ toss which means we ultimately tossed coin thrice.

Thus the probability will be $\frac{2}{4} = \frac{1}{2}$

$\therefore$ Required probability $= \frac{1}{2}$

From a group of 8 boys and 5 girls, a committee of 5 is to b

(5) From a group of 8 boys and 5 girls, a committee of 5 is to be formed. Find the probability that a committee contains

i) 3 boys and 2 girls ii) at least 3 boys.

Solution : A committee of 5 is to be formed

Total = 8B + 5G = 13

$\therefore n(S) = {}^{13}C_5 = \frac{13!}{8!\,5!} = 1287$

i) A : 3B 2G

$n(A) = {}^{8}C_3 \, {}^{5}C_2 = \frac{8!}{5!\,3!} \times \frac{5!}{3!\,2!} = 560$

$P(A) = \frac{n(A)}{n(S)} = \frac{560}{1287} = 0.43$

ii) B : At least 3 boys

$\therefore n(B) = {}^{8}C_3 \, {}^{5}C_2 + {}^{8}C_4 \, {}^{5}C_1 + {}^{8}C_5 \, {}^{5}C_0$

$= 56(10) + 70(5) + 56(1)$

$= 560 + 350 + 56 = 966$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{966}{1287} = 0.75$

(6) Letters in the word AHMEDNAGAR are rearranged. Find the probability that

(i) the arrangement begins and ends with A.

(ii) the arrangement begins with A but does not end with A

(iii) no two A's in the arrangement are adjacent to each other.

Solution : Since 10 letters are in the given word and A comes 3 times.

$\therefore n(S) = \frac{10!}{3!}$

i) event A = Arrangement begins and ends with A

ie. A X X X X X X X X A

$\therefore n(A) = 1 \times 8! \times 1 = 8!$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{8!}{(10!/3!)} = \frac{8! \times 3!}{10!} = \frac{1}{15}$

$\frac{(10!)(3!)}{10!} = \frac{1}{5}$

ii) B = Arrangement begins with A but does not end with A

$\therefore n(B) = 1 \times 7 \times \frac{8!}{2!}$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{1 \times 7 \times 8! \times 3!}{2! \times 10!} = \frac{7}{30}$

iii) C = No two A's in the arrangement are adjacent to each other.

ie. $\times H \times M \times E \times D \times N \times G \times R \times$

$\therefore n(C) = {}^{8}C_{3} \times 7! = \frac{8! \times 7!}{5! \times 3!}$

$\therefore P(C) = \frac{n(C)}{n(A)} = \frac{8! \times 7!}{3! 5!} \times \frac{3!}{10!} = \frac{7}{15}$

Sign of Teacher :